

Policy Trade-off Analysis

⚠ Historical Context Note: This analysis uses the **original index-based uniform P0** (deprecated), not the current spatially-stratified P0 baseline. The original P0 distributed units round-robin by firehouse index, resulting in poor geographic coverage (mean RT $\approx$ 18.5 min at K=20). The current P0 uses latitude-based spatial stratification and achieves 3.17 min mean RT at K=20. See [docs/nomenclature_migration.md](#) (nomenclature_migration.md) for the full nomenclature history and [docs/decisions_log.md](#) (decisions_log.md) DEC-011/DEC-012 for the rationale. The plots and tables below are retained for historical reference but should not be compared directly to production results that use the current P0 baseline.

Overview

The **Response Time vs Coverage Trade-off** plot ([fig_tradeoff_curve.png](#) and improved versions) is one of the key decision-support visualizations in this project. It compares all allocation policies across multiple fleet sizes on two performance dimensions simultaneously, revealing which policies are efficient and which are dominated.

What the Plot Shows

Axes

Axis	Metric	Definition
X-axis	Expected Response Time (minutes)	Demand-weighted average of the minimum travel time from the nearest active firehouse to each precinct. Computed as: $\text{sum}(\text{min_tt_j} * \text{demand_j})$ where min_tt_j is the travel time from the closest firehouse with ≥ 1 unit to precinct j . Lower is better.
Y-axis	% Demand Covered (≤ 8 min)	Percentage of total crash demand (weighted by precinct arrival rate) that falls within 8 minutes of travel time from an active firehouse. The 8-minute threshold follows the NFPA 1710 standard. A complementary 6-minute threshold aligns with NYC local law. Higher is better.

Visual Encodings

- **Color** encodes fleet size: K=20 (blue), K=30 (orange), K=40 (green), K=48 (red)
- **Marker shape** encodes policy (in improved version): P0=square, P1=triangle, P2=circle, P2b=diamond, P2c=plus
- **Ideal location**: upper-left corner (low response time, high coverage)

Policies Compared

ID	Policy	Type	Description
P0	Uniform Baseline	Non-optimized	Distributes K units as evenly as possible across all firehouses in the travel-time index. This is the standard uniform allocation (round-robin) – not the spatially-stratified P0-spatial variant.
P1	Demand-Proportional	Non-optimized	Allocates units proportional to crash demand of nearest precincts. A simple heuristic that concentrates units in high-demand areas.
P2	Demand-Weighted Optimized	MIP (PuLP/CBC)	Minimizes demand-weighted expected response time. Primary recommended policy.
P2b	P-Median Optimized	MIP (PuLP/CBC)	Selects K firehouses and assigns precincts to minimize total demand-weighted distance.
P2c	Maximal Coverage Optimized	MIP (PuLP/CBC)	Maximizes total demand covered within the 8-minute threshold.

Which P0 is Used?

P0 in this plot is the standard uniform allocation (`policies.uniform_allocation()`), which distributes K units round-robin across all 48 Manhattan firehouses. It is **not** the spatially-stratified P0-spatial (latitude/grid/maximin) introduced later in the project for fairer baseline comparison.

This matters because:

- With K=20, uniform allocation spreads 1 unit each across 20 of 48 firehouses (round-robin by index order), which can leave large geographic gaps, hence the poor performance (RT=18.5 min, cover-

age=73%).

- The spatially-stratified P0 would perform better at low K because it selects firehouses with geographic spread.

Key Findings from the Data

Performance Summary

K	P0 RT	P0 Cov	P1 RT	P1 Cov	P2 RT	P2 Cov	P2b RT	P2b Cov	P2c RT	P2c Cov
20	18.55	73.0 %	2.87	100%	2.54	100%	2.54	100%	3.48	100%
30	7.00	92.0 %	2.52	100%	2.49	100%	2.49	100%	3.48	100%
40	3.02	100%	2.52	100%	2.49	100%	2.49	100%	3.48	100%
48	2.49	100%	2.52	100%	2.49	100%	2.49	100%	3.48	100%

Pareto Frontier

The **Pareto frontier** (efficient frontier) consists of policies where you cannot improve one metric without worsening the other. In this dataset:

1. **P2 and P2b dominate all other policies** at every fleet size – they achieve the lowest response time while maintaining 100% coverage at all tested K values.
2. **P2 and P2b are essentially identical** in performance (both achieve RT=2.49 min at $K \geq 30$), meaning the demand-weighted and p-median formulations converge to the same solution for this problem geometry.
3. **P1 is near-efficient** – only ~0.4 min worse than P2 at $K=20$ (2.87 vs 2.54), making it a strong heuristic.
4. **P2c is consistently suboptimal** on the RT dimension (RT=3.48 at all K), because it optimizes for coverage threshold rather than minimizing response time.
5. **P0 is severely dominated at low K** ($K=20$: RT=18.5 min, 73% coverage) but converges to optimal at $K=48$ when every firehouse gets a unit.

Why Points Overlap

Most points for $K \geq 30$ cluster in the upper-left corner (RT ~ 2.5 min, coverage ~ 100%) because:

- Manhattan's 48 firehouses are dense enough that with 30+ units, all policies achieve near-full coverage
- The optimization models converge to similar solutions when capacity is not constraining
- The real differentiation happens at **low fleet sizes** ($K=20-30$), which is the operationally relevant range

Downstream Usage

Consumer	How it's used
Technical Report (Section 5)	Primary evidence for policy recommendation
Policy Recommendation (DEC-007)	Supports “P2 with K=20-30” recommendation
Fleet Sizing Decision	Shows diminishing returns above K=30
Stakeholder Communication	Visual summary of the cost-effectiveness trade-off

Files

File	Description
scripts/run_optimization_comparison.py (lines 253-273)	Original generation code
scripts/generate_tradeoff_improved.py	Improved visualization script
results/figures/fig_tradeoff_curve.png	Original plot (has overlapping labels)
results/figures/re- sponse_time_coverage_tradeoff_improved.png	Improved main plot with data table
results/figures/re- sponse_time_coverage_tradeoff_zoomed.png	Zoomed view of high-performance region
results/optimization/policy_comparison.csv	Source data

Recommendations Based on Trade-off

1. **Use P2 (Demand-Weighted) as the primary policy.** It dominates or ties all alternatives at every fleet size.
2. **K=20-30 is the sweet spot.** Beyond K=30, there is minimal improvement in either RT or coverage.
3. **P1 is a reasonable fallback** if optimization infrastructure is unavailable – it performs within 15% of optimal.
4. **P2c (Maximal Coverage) is not recommended** as the primary policy – it consistently underperforms P2 on response time while offering no coverage advantage.
5. **P0 (Uniform) should only be used as a comparison baseline**, not for operational deployment, especially at low K.